
0.78901
0.01234


0.90123
0.45678
0.34567

1.

Introduction to statistics

What is Statistics?

- is the science that deals with (Collecting, organizing, presenting, analyzing, and interpreting data to make decisions).

Branches of Statistics:

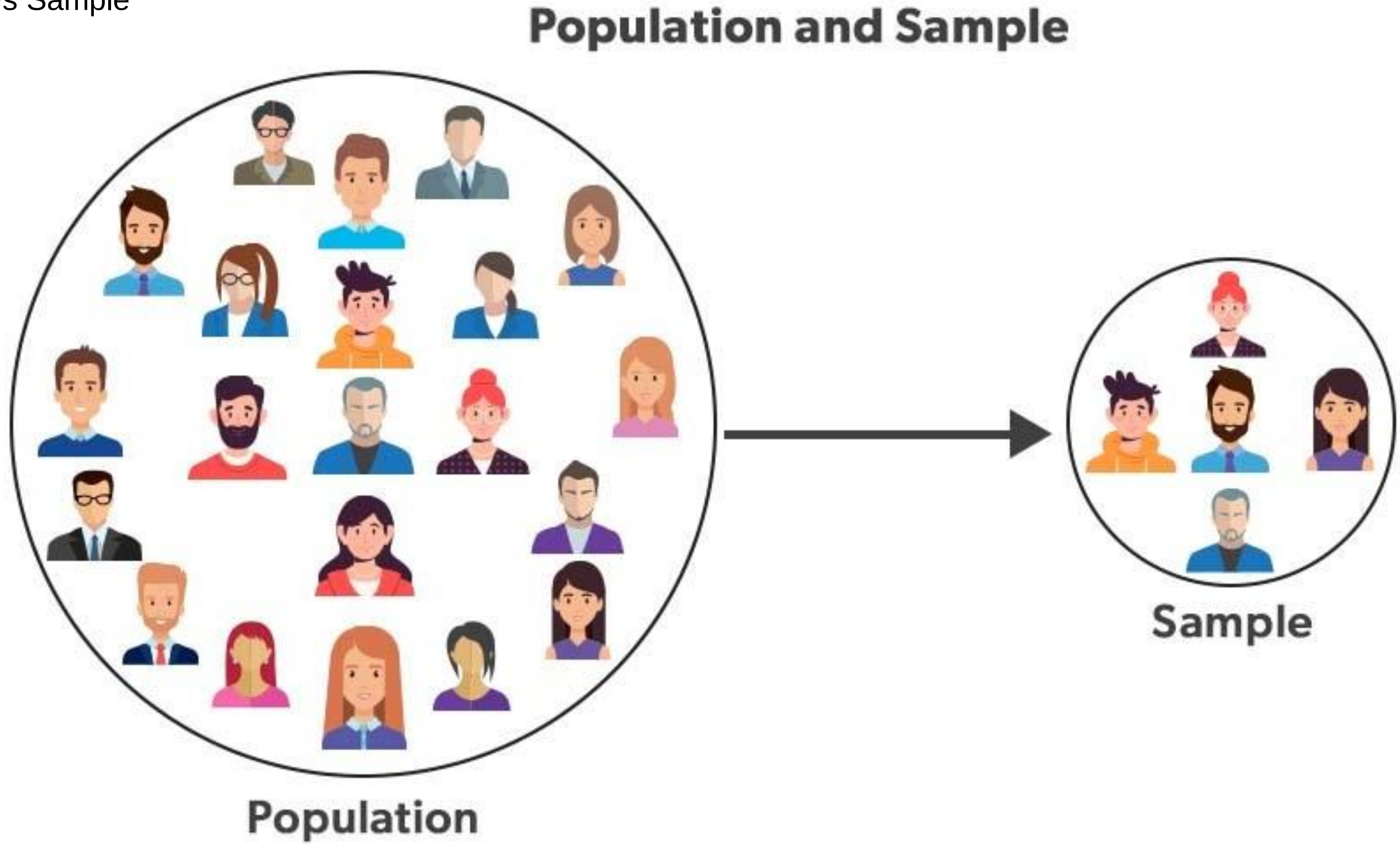
Descriptive Statistics

Deals with methods for organizing, displaying, and describing data by using tables, graphs, and summary measures.

Inferential Statistics

Deals with methods that use sample results to help make decisions or predictions about a population

Population vs Sample



Population Vs Sample

Population

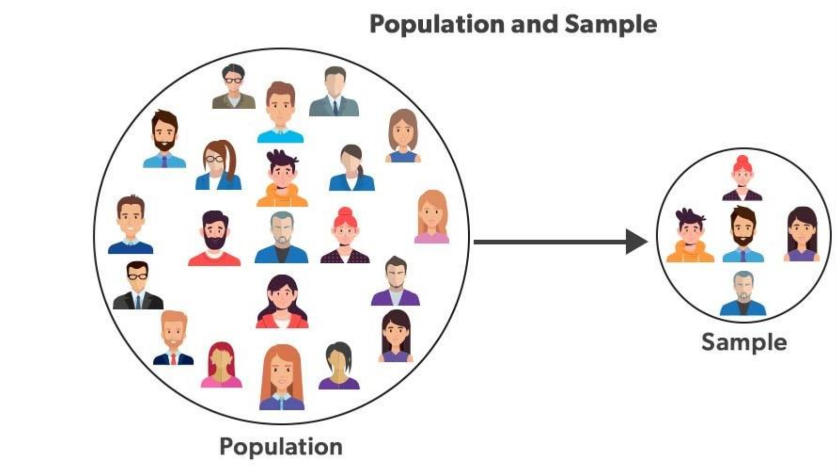
Consists of all elements–individuals, or objects-whose characteristics are being studied

Example :
All 5 million Florida voters who voted in today's election

Sample

A portion of the population selected for study is referred to as a sample.

Example : a sample of 1000 voters exit polls on election day.



Parameter Vs Statistics

Parameter

is the numerical description of population characteristics

-Usually the value of the population is unknown

- Population mean μ
- Population variance σ^2
- Population standard deviation σ

Statistics

is the numerical description of sample characteristics

- Statistic (or estimator) is an estimate of the population parameter

- Sample mean $\bar{x}$
- Sample variance S^2
- Sample standard deviation S

How to collect the data ?

Census

A survey that includes every member of the population is called a census.

- It gives actual and accurate results
- Often the target population is very large.
- In many cases, it is even impossible to identify each element of the population.
- Hence, in practice, a census is rarely taken because it is expensive and time-consuming.

Sample survey

The technique of collecting information from a portion of the population is called a sample survey.

- Collecting the data from some elements in the population
- Usually we use the sampling to save time and money . Also, the population may be destroyed.

Representative sample: Is a sample that represents the characteristics of the population as closely as possible.

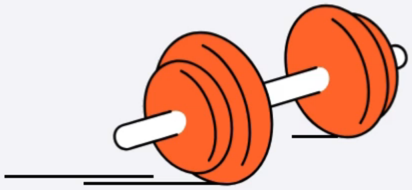
Random sample :

- A sample drawn in such a way that each element of the population has a chance of being selected is called a random sample.

- If all samples of the same size selected from a population have the same chance of being selected, we call it simple random sampling. Such a sample is called a simple random sample.

- A sample may be selected with or without replacement. In sampling with replacement, each time we select an element from the population, we put it back in the population before we select the next element. Sampling without replacement occurs when the selected element is not replaced in the population.

Recall: Data Types



Data Types

number of
hours of sleep

age group

amount of
distance walked

Classification of statistical data

- Types of variables
  - Quantitative (numerical)
    - Continuous
    - Discrete
  - Qualitative (categorical)
    - Ordinals
    - Nominals

Types of variables

QUANTITATIVE The variable can be measured numerically.		QUALITATIVE (categorical) can be classified into two non numeric categories	
Discrete	**Continuous**	**Nominal** **(non order متقدرش ترتبها)**	**Ordinal** **(can be ordered تقدر ترتبهم)**
The variable can assume certain values with no intermediate values. Examples: - No. of children, No. of cars, No. of students, No. of defective units ………	The variable can assume any numerical value over a certain interval. Examples: - Time, temperature, income, length, weights, age, and heights ……….	* Example: -The gender (M, F). -The colors, opinion of the people. - The marital status (d, m, s)……..	Example: - The grades are: V.W, W, G, V.G, and EX - Performance

REPRESENTING DATA

CLASS PRESENTATION

WHY ARE THEY USEFUL?

ORGANISE INFORMATION

COMPARE DATA

EASY TO UNDERSTAND AND SHARE

WHY ARE THEY USEFUL?

MAKE DECISIONS

EDUCATE OTHERS

SPOT TRENDS/ PATTERNS

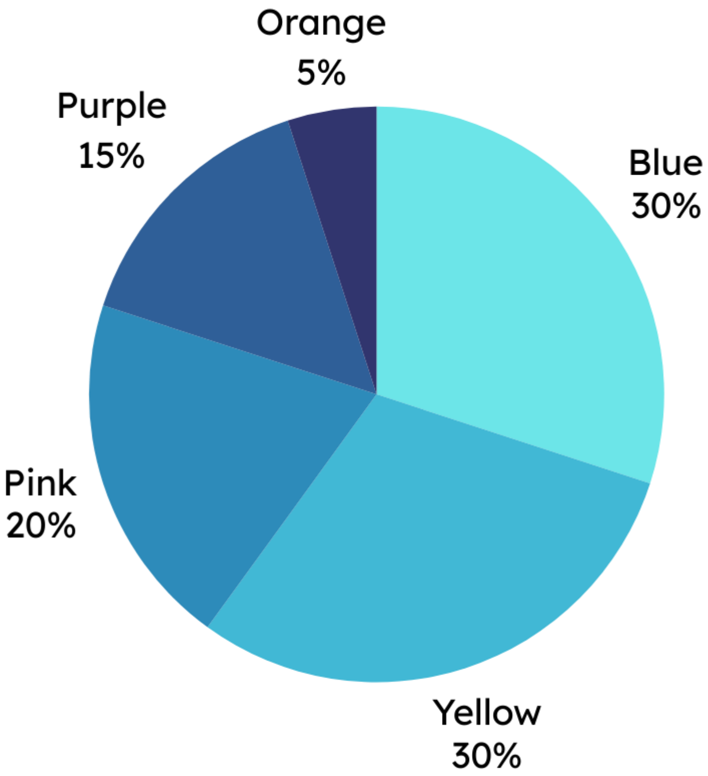
Let's explore different ways we can represent data!

TABLES

A table is a grid with rows and columns for organising data/information.

Use a table when you need to organise detailed numerical or categorical information.

Subject	Number of Students
History	19
Math	27
English	25
PE	18
Science	28



PIE CHART

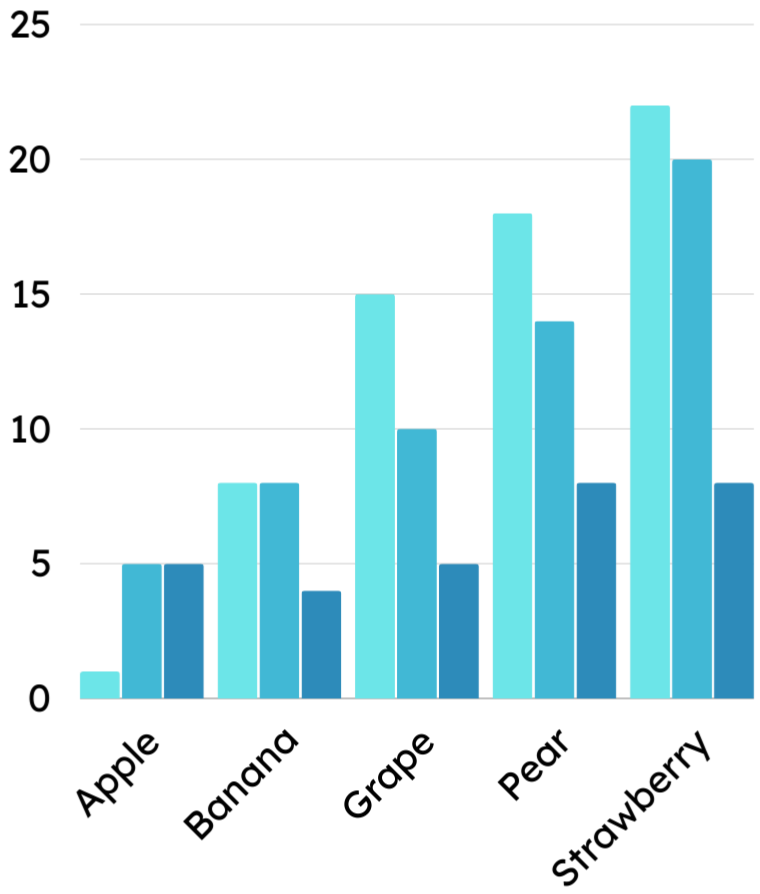
A pie chart looks like a pizza and divides up one whole into parts.

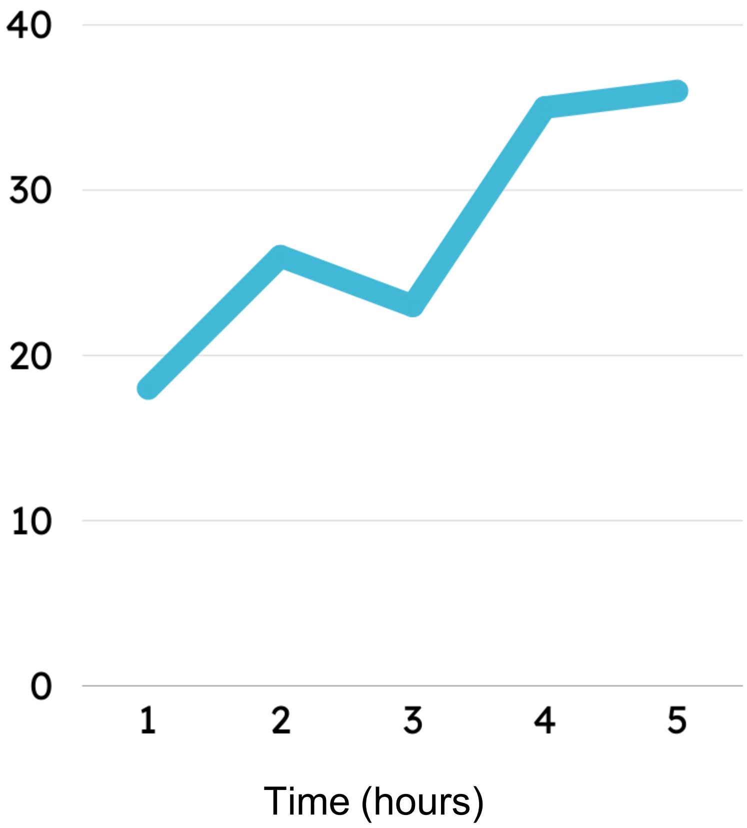
Use a pie chart when you want to show how something is divided into parts. They are great for data given in percentages!

BAR CHART

A bar chart uses bars (vertically or horizontally) to show and compare data.

Use a bar chart when you want to compare different categories or items.





LINE GRAPH

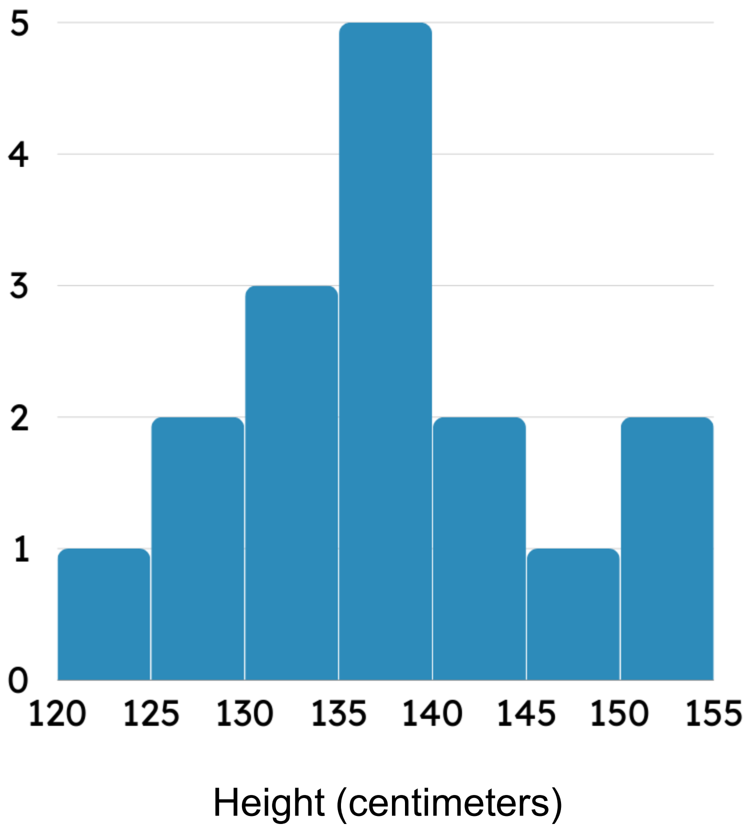
A line graph uses lines to show how data changes over time.

Use a line graph when you want to track how something changes over different timescales (minutes, hour, days etc.)

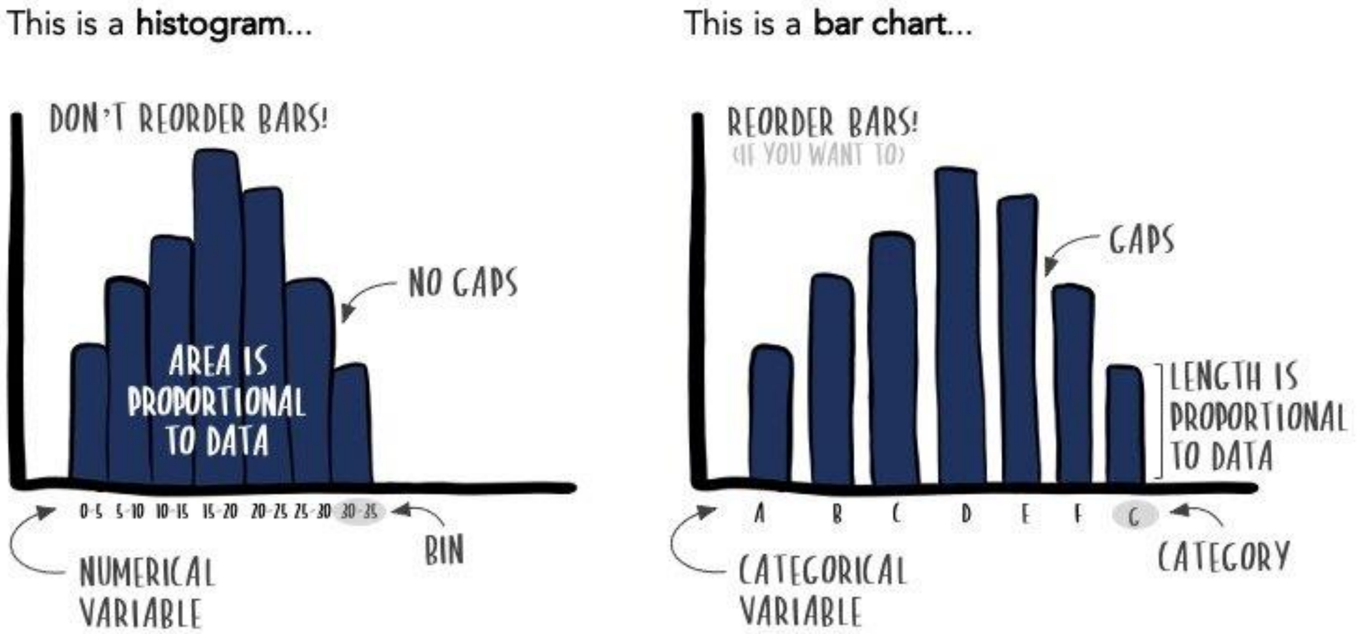
HISTOGRAM

A histogram is like a bar chart but for numbers in ranges (not categories).

Use a histogram when you have data in different ranges, like measuring the heights of people in your class.



What is the difference between a bar chart and a histogram ?


This is a histogram...
DON'T REORDER BARS!
AREA IS PROPORTIONAL TO DATA
NO GAPS
0-5 5-10 10-15 15-20 20-25 25-30 30-35
BIN
NUMERICAL VARIABLE
This is a bar chart...
REORDER BARS!
(IF YOU WANT TO)
GAPS
LENGTH IS PROPORTIONAL TO DATA
A B C D E F G
CATEGORY
CATEGORICAL VARIABLE

Histogram

Representing Data

Representing Data

Representing Data

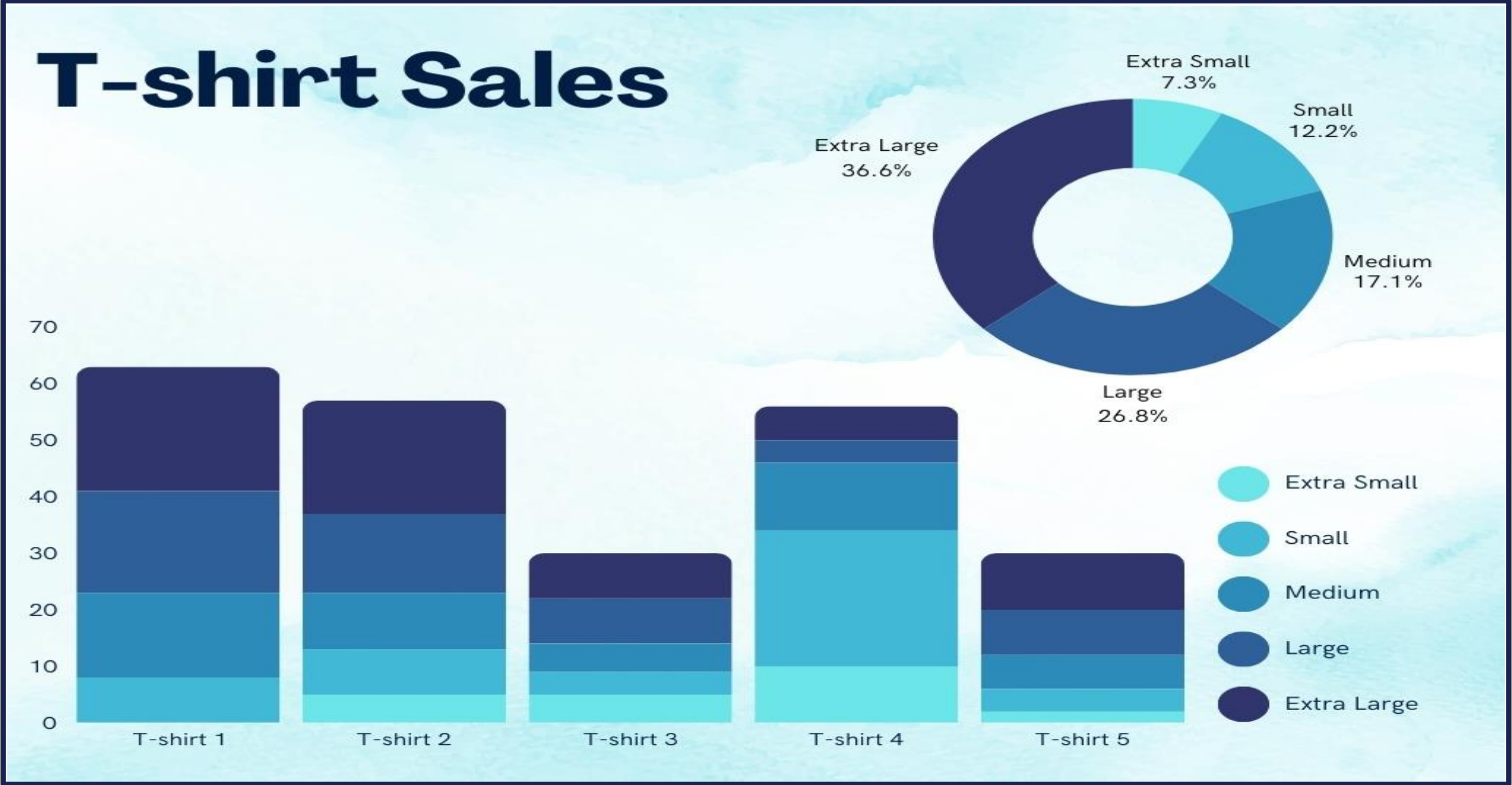
What did you understand from This table

Rank	Country/Economy	GDP (millions of $)	Agriculture GDP			Industry			Services			Year GDP	Year Sector
			GDP	Rank	% of GDP	GDP	Rank	% of GDP	GDP	Rank	% of GDP		
1	United States	19360000	174240	3	0.9	3659040	2	18.9	15526720	1	80.2	2017	2017
2	China	11940000	991020	1	8.3	4716300	1	39.5	6232680	2	52.2	2017	2017
3	Japan	4884000	48840	13	1	1450548	3	29.7	3384612	3	69.3	2017	2017
4	Germany	3652000	21912	31	0.6	1099252	4	30.1	2530836	4	69.3	2016	2017
5	France	2575000	51500	11	2	517575	7	20.1	2005925	6	77.9	2017	2017
6	United Kingdom	2565000	15390	36	0.6	487350	8	19	2062260	5	80.4	2017	2017
7	India	2439000	375606	2	15.4	560970	6	23	1499985	8	61.5	2017	2016
8	Brazil	2081000	129022	4	6.2	437010	12	21	1514968	7	72.8	2017	2017
9	Italy	1921000	40341	17	2.1	461040	10	24	1419619	9	73.9	2017	2017
10	Canada	1640000	27880	25	1.7	460840	11	28.1	1151280	10	70.2	2017	2017
11	Korea, South	1530000	33660	22	2.2	601290	5	39.3	891990	14	58.3	2017	2017
12	Russia	1469000	69043	7	4.7	475956	9	32.4	915187	13	62.3	2017	2017
13	Australia	1390000	50040	12	3.6	362790	14	26.1	977170	11	70.3	2017	2017
14	Spain	1307000	33982	21	2.6	303224	16	23.2	969794	12	74.2	2017	2017
15	Mexico	1142000	44538	15	3.9	360872	15	31.6	730880	15	64	2017	2017
16	Indonesia	923300	128339	5	13.9	372090	13	40.3	423795	19	45.9	2016	2017
17	Turkey	841200	56360	10	6.7	267502	18	31.8	516497	17	61.4	2017	2017
18	Netherlands	824500	13192	38	1.6	147586	27	17.9	578799	16	70.2	2017	2017
19	Saudi Arabia	683700	17776	32	2.6	302195	17	44.2	363728	22	53.2	2017	2017
20	Switzerland	680600	4764	77	0.7	174234	24	25.6	501602	18	73.7	2017	2017
21	Argentina	619900	67569	9	10.9	174812	23	28.2	377519	21	60.9	2017	2017
22	Taiwan	571500	10287	44	1.8	205740	19	36	354902	23	62.1	2017	2017
23	Sweden	541900	8670	50	1.6	178827	22	33	354403	24	65.4	2017	2017
24	Poland	510000	12240	41	2.4	205020	20	40.2	292740	26	57.4	2017	2017
25	Belgium	491700	3442	91	0.7	107191	32	21.8	381068	20	77.5	2017	2017
26	Thailand	437800	35900	18	8.2	158484	25	36.2	243417	30	55.6	2017	2017
27	Iran	427700	41915	16	9.8	153544	26	35.9	232241	34	54.3	2017	2017

Cross tables

What did you understand from This table

Year Sector	Sum of Agriculture GDP	Sum of Industry GDP	Average of Services GDP
2000	88	358	1429
2002	10	838	1668
2006	5	39	217
2008	1	56	247
2009	122	1263	2521.5
2010	114	137	2504
2011	12648	76136	35216
2012	323	1986	3043.666667
2013	4511	2464	2163.5
2014	467	2736	3469
2015	815	1879	2294.666667
2016	392414	582490	133156.4167
2017	3037077	21320941	287895.0773
FY12/13	68	883	5841
Grand Total	**3448663**	**21992206**	**250191.2651**

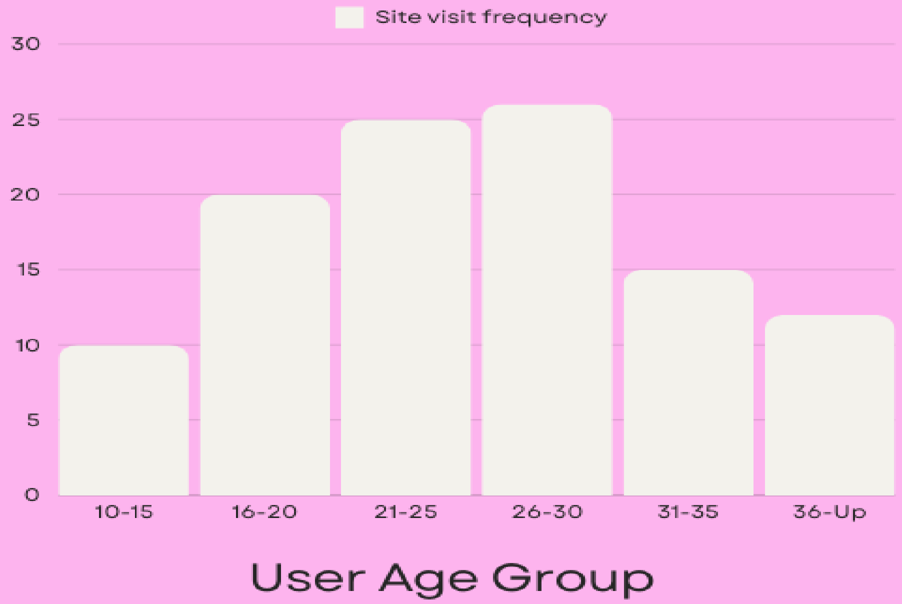

T-shirt Sales
Extra Small
7.3%
Small
12.2%
Medium
17.1%
Large
26.8%
Extra Large
36.6%
70
60
50
40
30
20
10
0
T-shirt 1
T-shirt 2
T-shirt 3
T-shirt 4
T-shirt 5
Extra Small
Small
Medium
Large
Extra Large

Monthly Website Activity

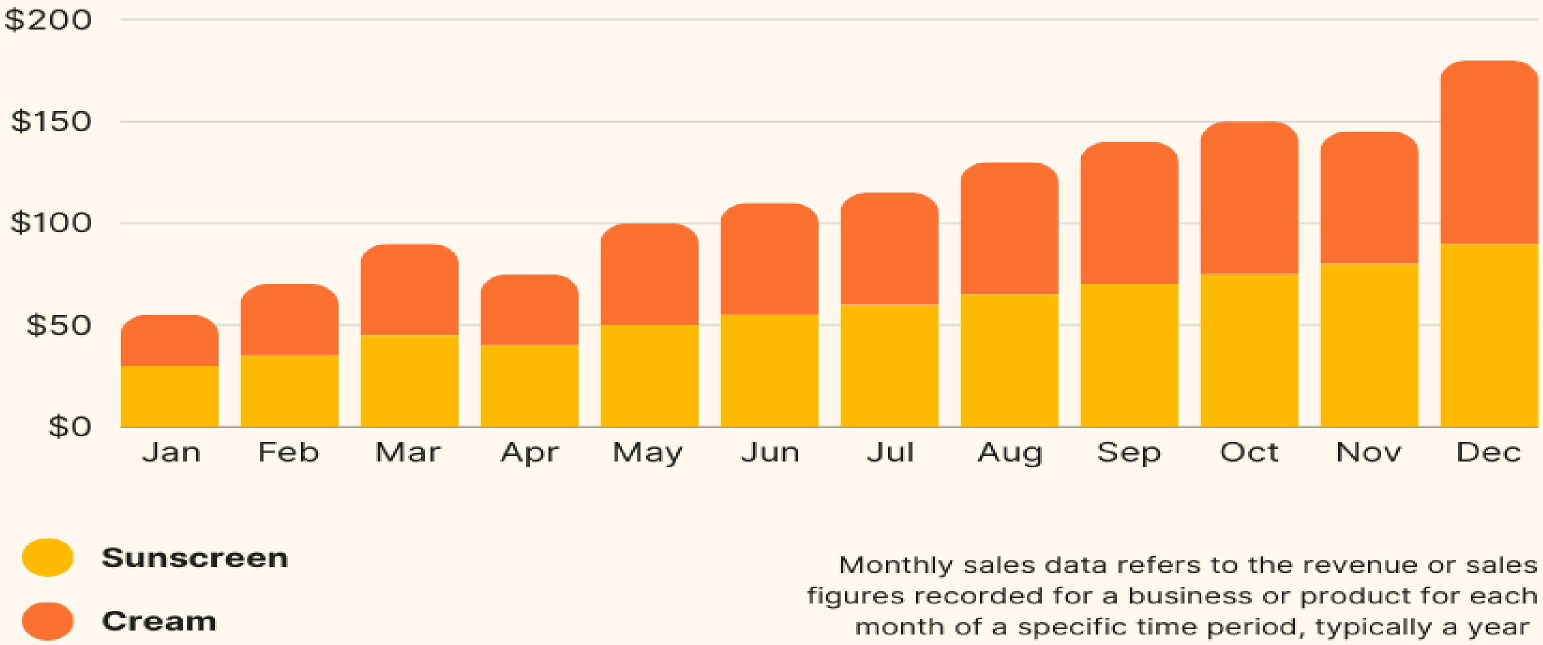
October 2031

An overview of which age groups have shown the most activity since the launch of the campaign

Campaign period:
June-August 2030



MONTHLY SALES DATA

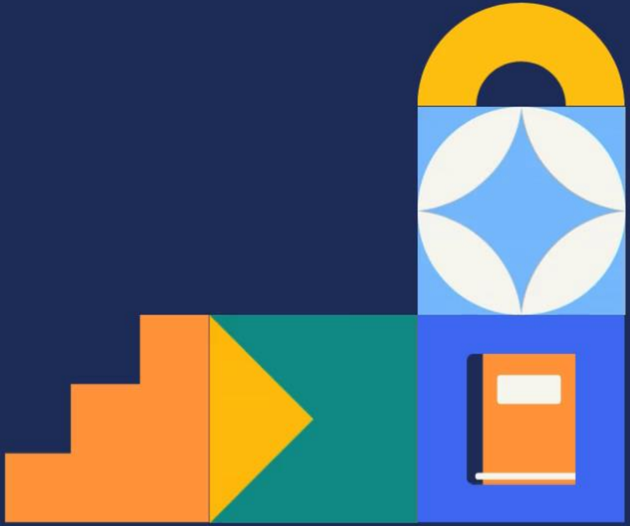


Monthly sales data refers to the revenue or sales figures recorded for a business or product for each month of a specific time period, typically a year

Design a survey for classmates. Choose one question from the provided options or create your own.

- What are your preferred sports?
- How many pets do you have at home?
- What is your mode of transportation when traveling to school?

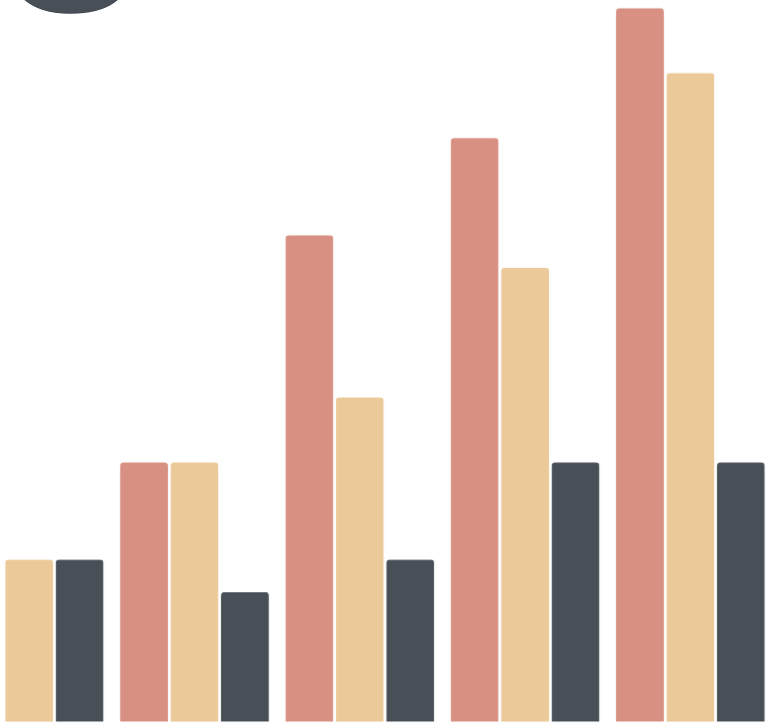
Organize, summarize, and present the data you collected in a poster.



NAME THE GRAPH

Statistics - Data Displays Quiz

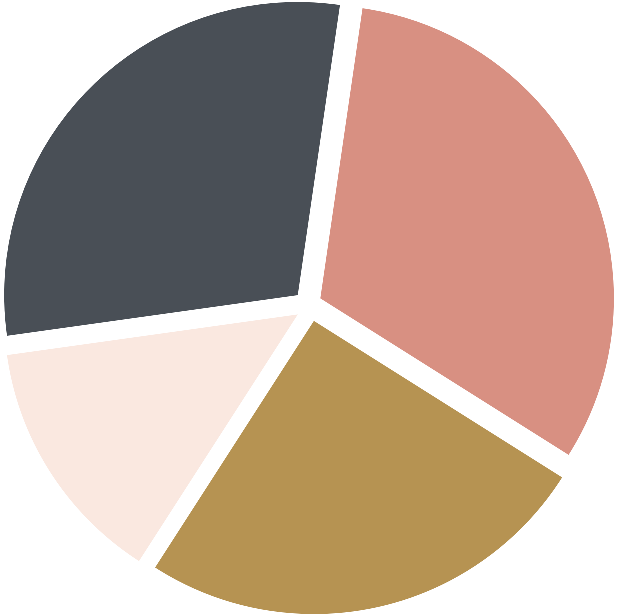
1



What is this graph called?

a. pie chart
b. dot plot
c. histogram
d. column chart

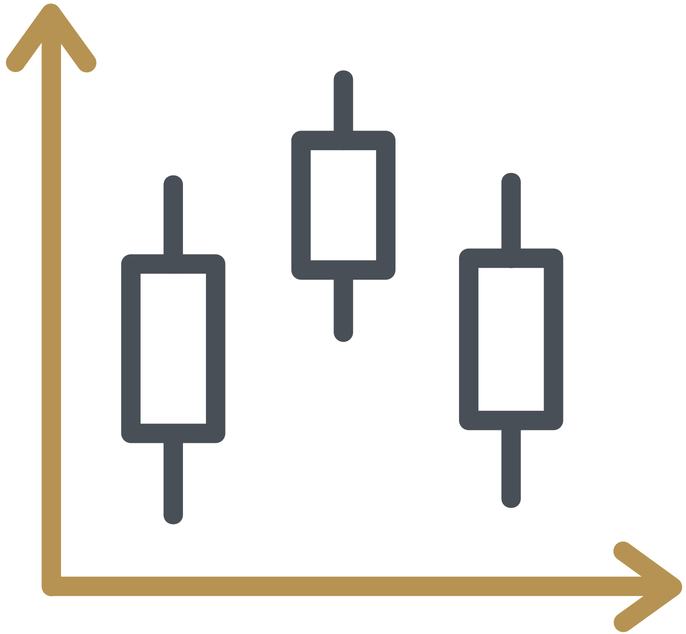
2



What is this graph called?

a. box & whisker plot
b. donut graph
c. pie chart
d. pictogram

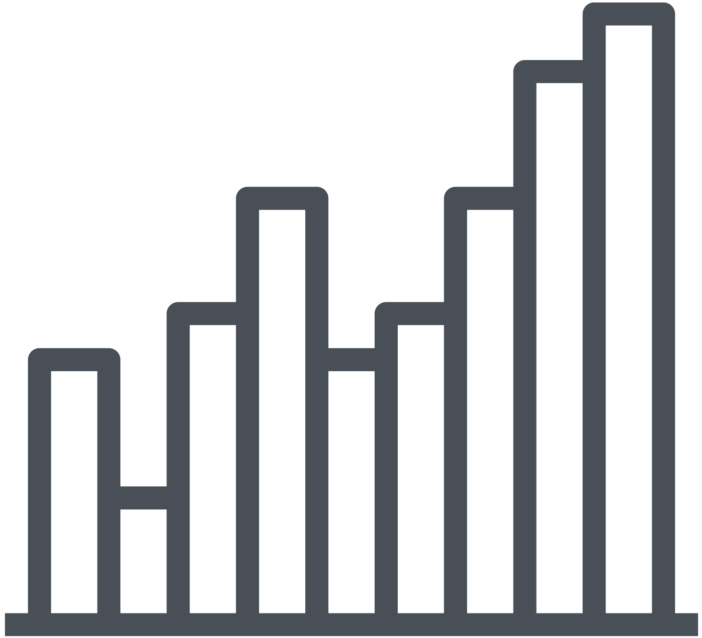
5



What is this graph called?

a. rectangle graph
b. bar chart
c. pie chart
d. box & whisker plot

6



What is this graph called?

a. bar chart
b. histogram
c. pie chart
d. dot plot

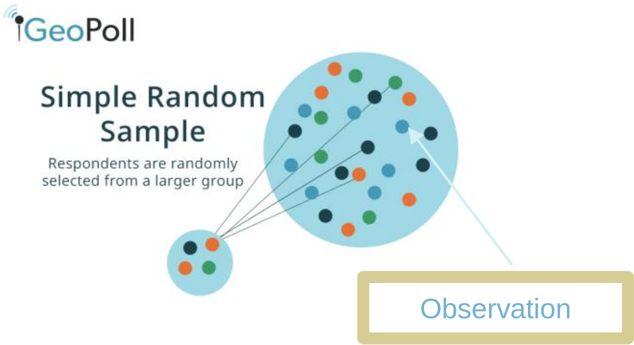
Sampling methods

Simple random sample

Stratified Sampling

Multistage sampling

Cluster sampling



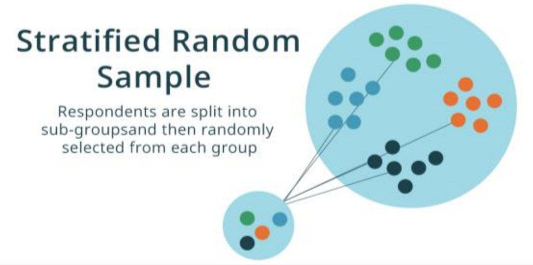
If we want to study the average yearly income of any human

Simple sample : In which each person has the same probability of existence in the sample

So, in this type of sampling we might get more Africans that Europeans because we are totally taking random samples .
We can get more males than females as well

So, there is no any kind of rule that govern the existence of any person in our sample

Simple sampling



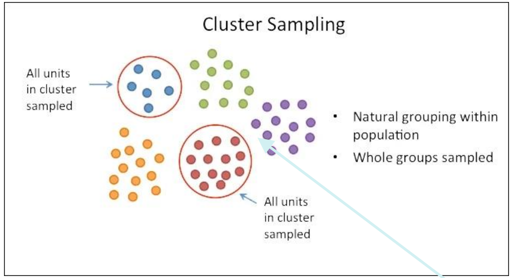
Observation

If we want to study the average yearly income of any human

Stratified sampling : is when we take random citizens from each country and study their yearly income

A **strata** is a group of homogenous observations

Stratified sampling

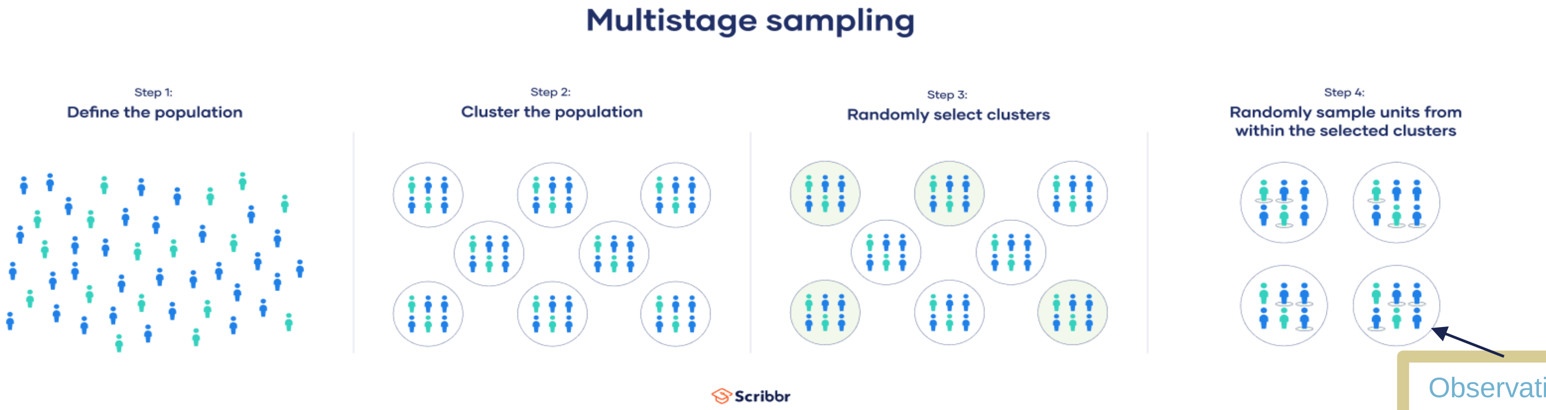


If we want to study the average yearly income of any human

In which we choose a city from each country and study their citizen (All of them)

Observation

Cluster sampling



Observation

If we want to study the average yearly income of any human

Sampling strategies

In which we, randomly, choose a city from each country and study only some of their randomly chosen citizens (Not all of them)

Multistage sampling

We choose our sampling strategy based on our population and our study

What type of sampling strategy to choose ?

Assume we want to study the effect of a new drug to sleeping hours , so what sampling strategy to choose ?

Let's assume we took a **simple** random sample , then this means that we might get women more than men or youths more than old people , so, can we take a **stratified** sample ?

A **stratified** random sample can be more representative to all

A **cluster** sample may not be the best sample to choose here because we would only study people of (55-60) , (15-20) and (40-55) then where are the others ? , we still have no statistics about them

And the **multistage** sample can not work as well

We can take a simple random sample if the population is homogeneous

Presenting Results

- Frequency
- Mode

Summarizing Results

- Mean
- Median

Summarize data using frequency, mode, mean, and median.

Present data using a frequency table.

Create a conclusion from the data.

Presenting Results

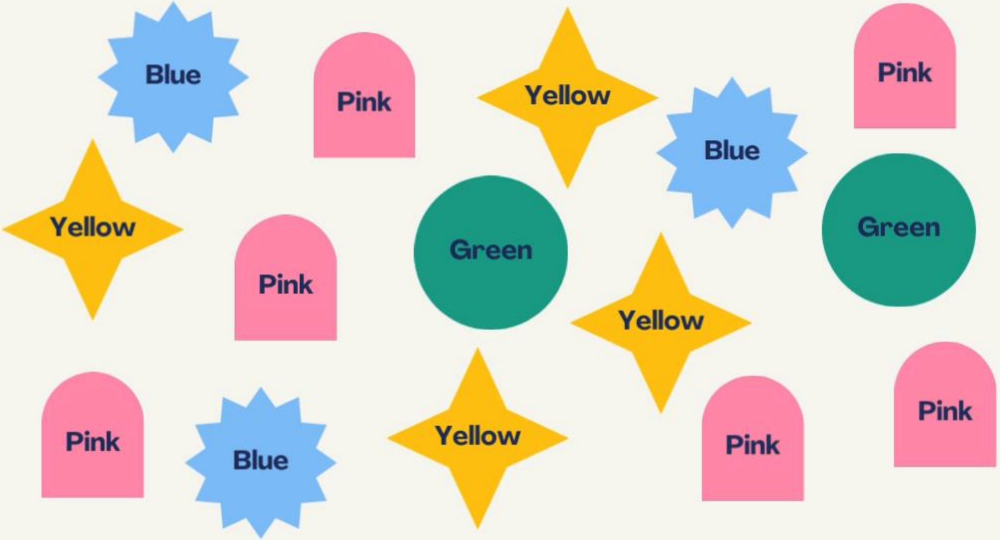
Frequency

refers to the number of times a result appears in the data set

Mode

refers to the result with the greatest frequency

Jan asks 15 of her friends what their favorite color is. Here are the answers she gathered:



Presenting Results

It is used to represent the frequency of different results in the data.

result or data value

frequency

Color	Number of Friends

Presenting Results

The equivalent frequency of Jan's poll is shown below:

Color	Number of Friends
Pink	6
Green	2
Blue	3
Yellow	4

Presenting Results

The equivalent frequency of Jan's poll is shown below:

Color	Number of Friends
Pink	6
Green	2
Blue	3
Yellow	4

A grade 6 class has the following sets of scores in a quiz. Create a frequency table for the data set.

14, 2, 5, 2, 2, 5, 5, 7, 12, 13, 13, 14

1. What is/are the mode/s of the data set?
2. How many students are there in the class?

Score	Frequency

Answer Key

1. The modes of the data set are 2 and 5.
2. There are 12 students in the class.

modes:
2 and 5

Score	Frequency
2	3
5	3
7	1
12	1
13	2
14	2

Total: 12

Measures of Central Tendency

EXAMPLE

Consider the data of 5 runners below.

What is the mean, median and mode?

Respondents	A	B	C	D	E
Distance Ran (km)	8	8	10	7.5	8

FORMULAE OF THE MEAN

$$\textbf{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

$$Mean = \frac{8 + 8 + 10 + 7.5 + 8}{5}$$

$$Mean = \boxed{8.3 \text{ km}}$$

EXAMPLE

What is the mean, median and mode?

MEDIAN

Step 1: Arrange the values in order.

7.5, 8, 8, 8, 10

Step 2: Identify the middlemost value.

7.5, 8, 8, 8, 10

MEDIAN = 8

MODE

Identify the value that occurs the most.

7.5, 8, 8, 8, 10

MODE = 8

Summarizing Results

Find the median of the following data sets:

Set A: 8, 3, 4, 9, 6

3, 4, 6, 8, 9

$\downarrow$

Median

Set B: 11, 17, 3, 14, 19, 7

3, 7, 11, 14, 17, 19

Median = $\dfrac{11 + 14}{2}$ = 12.5

Which two data sets have the same mean?

A 9, 11, 15, 21, 7

B 12, 11, 14, 15, 9

C 7, 11, 30, 8, 9

D 6, 14, 10, 13, 22

Find also the median for all four data sets.

Measures of Central Tendency

MEAN

average value of the data

MEDIAN

middlemost value of the data

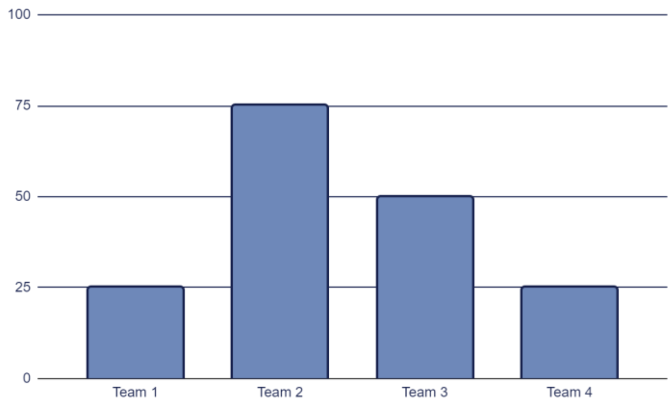
MODE

most frequent value of the data

Find the mean, median and mode of the line graph.

Share your answer in our next session.

Choose your team!



Look at the graphic and **calculate the arithmetic mean and the distribution range** of the score obtained by the following soccer teams.

The following frequency distribution represents the weekly demand on one product of a
company over the past 50 weeks. Compute the *average* size of the weekly demand on this product.

# demanded units (x)	5	6	7	8	9	10	Total
# weeks (f)	8	12	15	8	5	2	50

Solution:

$$\text{Average} = \frac{\sum xf}{\sum f} = \frac{346}{50} = 6.92$$

x	f	$x.f$
5	8	40
6	12	72
7	15	105
8	8	64
9	5	45
10	2	20
Total	50	346

The following data set represented the Math.

Scores of a random sample of 100 students in the English on the first year:

Class	30-	40-	50-	60-	70-80	Total
Frequency	10	20	40	20	10	100

Find the Arithmetic Mean.

Solution:

class	F	X (Midpoint)	X.F
30-	10	$\frac{30+40}{2}=35$	$10\times35=350$
40-	20	$\frac{40+50}{2}=45$	$20\times45=900$
50-	40	$\frac{50+60}{2}=55$	$40\times55=2200$
60-	20	$\frac{60+70}{2}=65$	1300
70-80	10	$\frac{70+80}{2}=75$	750
total	100		5500

Arithmetic Mean $=\frac{\sum xf}{\sum f}=\frac{5500}{100}=55$

Properties of the Mean:

- A set of data has only one mean. The mean is unique.
- The mean is sensitive to extreme values (outliers); very large or very small values.
- Can be computed for quantitative data only.
- All the values are included in computing the mean.
- The sum of the deviations from the mean equals zero. Expressed as follow:

$$\sum(X - \bar{X}) = 0$$

Disadvantages:

- It's affected by extremes (outliers)
- we can't compute it for qualitative data

Measures of Central Tendency	Mean	Median	Mode
Computed for	Quantitative data (only)	Quantitative data + Ordinal (ينفع تترتب)	Quantitative data + Qualitative data (All types of data)
Values for computing	All values	Not all values	The most frequently Repeated value
Sensitivity to outliers	Sensitive	insensitive	insensitive
Uniqueness	unique	unique	May exist one mode or two or more (or no mode)
	$\sum(x - \bar{x}) = 0$ The sum of the deviations from the mean equals zero		

Try This!

Use the following frequency distribution for questions 1-3:

The frequency distribution table of the daily wages for a random sample of 20 workers in the drug industry is given as:

Class	15-	25-	35-	45-	55-	65-75	Total
Frequency(f)	6	2	5	3	2	2	20

1. **The number of workers whose wages ranges from 25 up to 65 equals:**
   a) 13 (b) 12 (c) 15 (d) 16

2. **The percentage of workers whose wages greater than or equal 35 is:**
   a) 60% (b) 65% (c) 85% (d) 80%

3. **The mean wage equals:**
   a) 39.5 (b) 40.5 (c) 44 (d) 45.5

4. If the average depth of a lake is 1.5 meters, it means that .....
   a) The deepest point of the lake is 1.5 meters.
   b) An adult of average height can walk through the lake.
   c) 50% of the lake has depth less than or equal to 1.5 meters.
   d) There could be a spot in the lake where it is deeper than 1.5 meters.

Measures of Dispersion

Measures of dispersion are descriptive statistics that describe how similar a set of scores are to each other

The more similar the scores are to each other, **the lower** the measure of dispersion will be

The less similar the scores are to each other, **the higher** the measure of dispersion will be

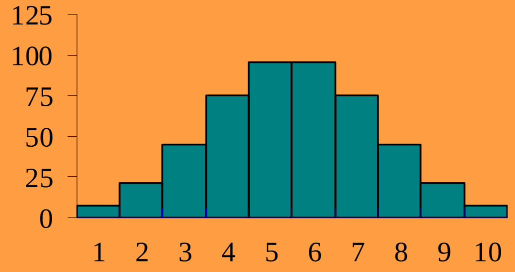
In general, the more spread out a distribution is, the larger the measure of dispersion will be

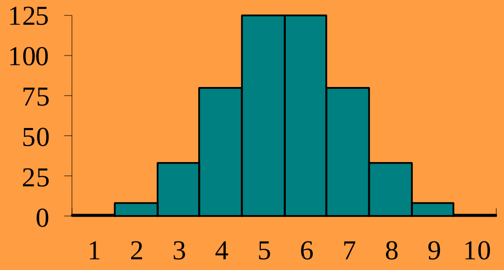
Note that all measures of dispersion are **nonnegative**. Also, a measure of dispersion takes on the value **zero only if** all observations in a data set have **the same value**.

Measure of **"absolute dispersion"** are used to describe the variability structure of just "one" group,
while measures of **"relative dispersion or coefficients of variation"** are used to compare the dispersion between "two or more groups".

Measures of Dispersion

- Which of the distributions of scores has the larger dispersion?





- The upper distribution has more dispersion because the scores are more spread out
  - That is, they are less similar to each other

Measures of Dispersion

- There are three main measures of dispersion:
  1. The range
  2. The Interquartile range (IQR)
  3. Variance / standard deviation

1) The Range

- The range is the simplest measure of dispersion to calculate. It is obtained by taking the difference between the largest and the smallest values in a data set.

  $X_L – X_S$

- What is the range of the following data:
  4 8 1 6 6 2 9 3 6 9

- The largest score (XL) is 9 ; the smallest score (XS) is 1;

- the range is $X_L – X_S = 9 - 1 = 8$

Properties of the range:

1) Very simple to compute.

2) Its calculation is based on only two values, the largest and the smallest, and these two values may be outliers or extremes.

3) Sensitive to extreme values.

2) The Interquartile Range

- The I.Q.R. measures the range of only 50% of the observations at the middle and it eliminates any information about the first and last quarters of the data.

- The first quartile is the 25th percentile

  - The third quartile is the 75th percentile
- IQR = $(Q_3 - Q_1)$

Properties of the Interquartile range:

<u>Properties of I.Q.R:</u>

1) **Depends on only 50% of the data.**

2) **Insensitive to extreme values or outliers.**

<u>Note that:</u>

Each of the range and the I.Q.R. depends on only two values, the smallest and the largest and, Q1 and Q3 respectively. A measure of dispersion, which makes a full use of the information provided by the data, will certainly be better.

3) Standard deviation and Variance:

$$\boldsymbol{Standard\ deviation} = \sqrt{\boldsymbol{Variance}}$$

- The standard deviation is the most used measure of dispersion.
- The value of the standard deviation tells how closely the values of a data set are clustered around the mean.
- The standard deviation is obtained by taking the positive square root of the variance.
- The variance calculated for population data is denoted byσ^2, and the variance calculated for sample data is denoted by S^2.
- Consequently, the standard deviation calculated data is denoted by $\boldsymbol{\sigma}$, and the standard deviation calculated for sample data is denoted by $\boldsymbol{S}$.

The formulas for the sample variance S^2 are:

Ungrouped	Grouped
$\dfrac{\sum (x - \bar{x})^2}{n - 1}$	$\dfrac{\sum f\,(x - \bar{x})^2}{\sum f - 1}$
$\dfrac{\sum x^2 - \dfrac{(\sum x)^2}{n}}{n - 1}$	$\dfrac{\sum x^2 f - \dfrac{(\sum x\,f)^2}{\sum f}}{\sum f - 1}$

Example (1):

Find the sample standard deviation from the following data:
4, 9, 5

Solution:

x	x - $\bar{x}$	$(x-\bar{x})^2$	x^2
4	4 – 6 = -2	4	16
9	9 – 6 = 3	9	81
5	5 – 6 = -1	1	25
$\sum x = 18$	0	$\sum(x-\bar{x})^2=14$	$\sum x^2 = 122$

$$\bar{x} = \frac{4+9+5}{3} = 6$$

The sample variance

$$s^2 = \frac{\sum(x - \bar{x})^2}{n-1} = \frac{14}{3-1} = \mathbf{7}$$

$$s^2 = \frac{\sum x^2 - \frac{(\sum x)^2}{n}}{n-1} = \frac{122 - \frac{18^2}{3}}{3-1} = \mathbf{7}$$

The sample standard deviation

$$S = \sqrt{7} = 2.645$$

Example (2):

The following data set represents the number of daily smoked cigarettes in a random sample of the Eng. Dept. students in the faculty of commerce

No. of daily smoked cigarettes	4	5	6	7	8	9
Frequency	4	10	20	25	30	11

Compute the standard deviation

Solution:

x	f	xf	$(x - \bar{x}($	$(x - \bar{x})^2$	$(x - \bar{x})^2 f$	$x^2 . f$
4	4	16	-3	9	36	64
5	10	50	-2	4	40	250
6	20	120	-1	1	20	720
7	25	175	0	0	0	1225
8	30	240	1	1	30	1920
9	11	99	2	4	44	891
Total	100	700			170	5070

$$\bar{x} = \frac{\sum xf}{\sum f} = \frac{700}{100} = 7$$

$$S^2 = \frac{\sum f\,(x - \bar{x})^2}{\sum f - 1} = \frac{170}{100 - 1} = 1.717$$

$$\text{S} = \sqrt{1.717} = 1.310$$

<u>Another solution:</u>

$$\boldsymbol{s^2} = \frac{\sum x^2 f - \frac{(\sum xf)^2}{\sum f}}{\sum f - 1} = \frac{5070 - \frac{700^2}{100}}{100 - 1} = 1.717$$

The sample standard deviation

$$\text{S} = \sqrt{1.717} = 1.310$$

Try This!

1

The following data set represents a random sample of the monthly income (in thousands of L.E) (40 , 3 ,3 ,4, 4, 6)

Find:

1) The range **2) The variance** **3) The standard deviation**

2

<u>Choose :</u>

If the statistics score of all students in some exam is 90 , this means that

The mode = zero b) The mode = 90

c) The Range= 90 d) The standard deviation = zero

3

In a data set, if all values are negative, then: .....

a)All measures of central tendency are negative, and all measures of dispersion are negative.

b)All measures of central tendency are positive, and all measures of dispersion are negative.

c)All measures of central tendency are negative, while all measures of dispersion are non-negative.

d)None of the above.

4

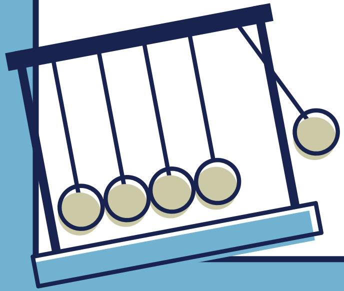
If Range(depth of an artificial lake) = 0 and median(depth) = 1 meter, this implies that

a)mean (depth) = 0 b)Standard deviation (depth) = 1

c)The sum of the deviations from 1 equals zero d)Mode (depth) = 0

“الإحصاء و الرياضيات زي الدنيا في بعض الأحيان مبتتعلمش غير لما تحاول تمشي صح بس تغلط , فتحاول تاني فتقع تاني بعدها بقى تتعلم من غلطاتك و كل مزادتك تجاربك و محاولة تعلمك من اللي فات كل لما إكتسبت خبرة”.

-أكمل العراقي



Thanks!

Do you have any questions?

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